

Communications — Mededelingen

A CARD-AGGLUTINATION TEST WITH STAINED TRYPANOSOMES (C. A. T. T.) FOR THE SEROLOGICAL DIAGNOSIS OF *T. B. GAMBIENSE* TRYPANOSOMIASIS

by

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Summary — A card-agglutination test (C. A. T. T.) has been developed for the serological diagnosis of *T. b. gambiense* trypanosomiasis. The antigen consists of a suspension of fixed and stained bloodstream trypanosomes of defined variable antigen type. The method has been evaluated by testing sera from 155 patients with proven *T. b. gambiense* infection, sera from 247 individuals free of sleeping sickness and sera from 54 patients with various parasitoses.

KEYWORDS : Sleeping Sickness; Serodiagnosis; *Trypanosoma brucei*; *Trypanosoma brucei gambiense*; Antigenic Variation; Direct Agglutination.

Introduction

Bloodstream trypanosomes of *Trypanosoma brucei* contain both stable (common, somatic) and variable (surface) antigens (Weitz, 1970; Seed, 1974; Le Ray, 1975). Relapsing infections result in the production of antibodies to both kinds of antigens (De Raadt, 1974). Antibodies to variable antigens can be selectively demonstrated by trypanolysis, neutralization, direct agglutination and precipitation tests (Gray and Luckins, 1976) and, in well-defined conditions, by immuno-enzymatic and immunofluorescence techniques too.

Recent studies on variable antigen type (VAT) repertoires revealed that certain variable antigens were produced in common by different strains, subspecies and species of the subgenus *Trypanozoon* (Gray and Luckins, 1976; Van Meirvenne *et al.*, 1977). During the present experiments a widespread predominant VAT was used to develop a direct agglutination test for serodiagnosis of *T. b. gambiense* trypanosomiasis.

Materials and Methods

Sera

Three main groups of human sera were examined :

Group 1 consisted of 247 sera collected from 56 Africans and 191 Caucasians free of trypanosomiasis. Group 2 included sera of 54 patients suffering from various other parasitoses (Table 3). The third group of sera had been collected before, during or after drug-treatment from 155 trypanosomized

individuals in endemic areas of sleeping sickness in Zaire (Mushie/Boma), Ivory Coast (Daloa/Bouaflé) and Upper Volta. Infections were confirmed by demonstration of trypanosomes either in blood or in lymph fluid. A detailed history of the serum samples of group 3 is given in tables 4, 5, 6 and 7.

Monospecific antisera against VATs AnTat 1 to 13 were raised in rabbits by intravenous inoculation of 10^6 clone trypanosomes and bleeding on the 6th day of infection (Van Meirvenne *et al.*, 1975).

Antigen preparation

A clone population of the VAT AnTat 8 derived from a stock of *T. b. brucei* (Van Meirvenne *et al.*, 1975, 1977) was used to prepare the antigen.

Rats were inoculated intraperitoneally with about 50×10^6 trypanosomes using parasitized mouse blood. Three days later, trypanosomes were harvested from the heavily infected blood by Lanham's column technique, using a phosphate-saline-glucose (PSG) 6:4 buffer system (Lanham and Godfrey, 1970).

Trypanosomes were spun down, resuspended in PSG and washed 3 times by centrifugation (4 °C, 15 min, 1,400 g). The sediment was resuspended in cold PSG so as to obtain a concentration of approximately 200×10^6 organisms per ml. This suspension was mixed with an equal volume of a cold formaldehyde solution (2 per cent w/v in PSG) and left at 4 °C for 20 hours.

Excess formaldehyde was removed by centrifugation and by two washes with cold phosphate-buffered saline (PBS, pH 7.2, 4 °C). Further operations were carried out at room temperature. A suspension in PBS containing about 200×10^6 trypanosomes per ml was prepared and added to an equal volume of a Coomassie Blue solution¹ (1:5,000 w/v in PBS). The suspension was gently mixed on a magnetic stirrer for 1 hour. After centrifugation and two final washes with PBS, a suspension in PBS containing approximately 100×10^6 trypanosomes per ml was prepared, filtered through nylon gauze (mesh size 40 μ), dispensed per 1, 3 or 5 ml into penicillin vials, snapfrozen in liquid nitrogen and stored at - 80 °C.

Test procedure

The test was performed on plastic-coated cards with ten « 18 mm circle » spots². One drop (50 μ l) of serum or twofold dilutions in PBS and one drop (50 μ l) of antigen were set out onto the reaction areas. The mixtures were spread out and allowed to react for two minutes on a specially designed rotator³. In a positive reaction the trypanosomes agglutinated as coarse blue particles which were macroscopically visible. The results were recorded as negative (-), weakly positive (+), positive (++) and strongly positive (+++) (see figure 1). The highest serum dilution still giving a + result was considered as the end-titre.

(1) Coomassie Brilliant Blue R-250, BioRad.

(2) Brewer Diagnostic Cards for R. P. R. No. 8718-49, Hynson, Westcott and Dunning Inc. Baltimore, Md. 21201.

(3) Card Test Rotator, Model 46, Hynson, Westcott and Dunning.

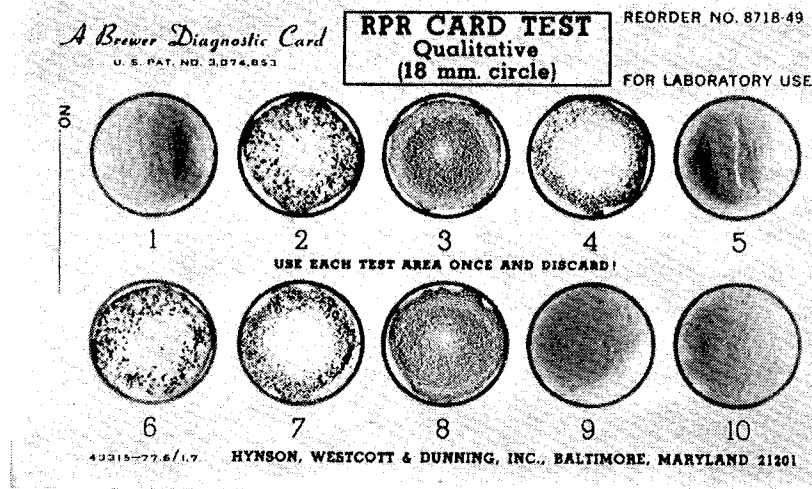


Figure 1

Examples of C. A. T. T. results.
 Spots 1 to 5 : screening of 5 sera (dilution 1 : 4).
 (1) : -, (2) : + + +, (3) : +, (4) : + +, (5) : -.
 Spots 6 to 10 : titration of a positive serum (dilutions 1 : 4 to 1 : 64).
 (6) : + + +, (7) : + +, (8) : + (endpoint), (9) : -, (10) : -.

Results

The VAT-specific character of the C. A. T. T. was demonstrated by titration of monospecific rabbit antisera against VATs AnTat 1 to 13. As may be seen from table 1, the end-titre of heterologous antisera never exceeded 1 : 4 whereas it was 1 : 256 for the homologous antiserum.

TABLE 1
 End-titres of VAT-specific rabbit antisera

VAT-specificity	End-titre
AnTat 1	Nonreactive
2	1 : 1
3	1 : 2
4	Nonreactive
5	1 : 2
6	1 : 1
7	1 : 2
8 *	1 : 256
9	Nonreactive
10	1 : 1
11	1 : 4
12	1 : 1
13	1 : 1

* Homologous combination.

In order to determine the specific threshold dilution, sera from groups 1 and 2 were titrated. Respectively 1.6 and 9.3 per cent of the samples gave a non-specific reaction up to the dilution 1 : 2 (Tables 2 and 3). It was concluded that the specific threshold dilution was 1 : 4.

TABLE 2

End-titre ranges of 247 sera from 56 Africans and 191 Caucasians free of trypanosomiasis (group 1)

Sera		End-titres			
Origin	Number	Nonreactive	1 : 1	1 : 2	1 : 4
Africans *	56	48	4	4	0
Caucasians	191	188	3	0	0
Total	247	236	7	4	0
	%	95.6	2.8	1.6	0

* Including 41 sera from Zairians.

TABLE 3

End-titre ranges of sera collected from 54 patients with various parasitoses (group 2)

Sera		End-titres			
Parasite	Number	Nonreactive	1 : 1	1 : 2	1 : 4
<i>T. gondii</i>	10	7	3	0	0
<i>Leishmania</i> sp.	5	3	2	0	0
<i>E. histolytica</i>	8	3	3	2	0
<i>T. cruzi</i>	2	1	1	0	0
<i>Plasmodium</i> sp.	3	3	0	0	0
<i>G. intestinalis</i>	5	5	0	0	0
<i>Filaria</i> sp.	5	3	1	1	0
<i>S. mansoni</i>	7	5	1	1	0
<i>Rickettsia</i> sp.	9	7	1	1	0
Total	54	37	12	5	0
	%	68.5	22.2	9.3	0

The sensitivity of the test was checked by titration of 155 sera from patients with active or cured *T. b. gambiense* infection.

A first series of sera had been collected at Mushie (Zaire) from 76 patients before or within 2 months following chemotherapy. All sera yielded positive results with end-titres ranging from 1 : 8 to 1 : 64 (Table 4).

TABLE 4

End-titre ranges of sera from 76 patients with *T. b. gambiense* (Mushie/Zaire)

Sera		End-titres				
Drug-treated since	Number	1 : 4	1 : 8	1 : 16	1 : 32	1 : 64
Untreated	19	0	5	7	6	1
1 to 7 days	33	0	7	17	5	4
1 to 4 weeks	15	0	4	6	4	1
1 to 2 months	9	0	2	5	2	0
Total	76	0	18	35	17	6

A second series of sera had been collected from 25 Zairians living in the area of Boma/Vindu. Four patients had received no trypanocidal drugs. The sera of two of them showed a titre of 1 : 8. Two samples gave a negative result but were also negative when tested by indirect immunofluorescence. Twenty-one patients had been treated with Arsobal since 4 to 9 months. Fourteen sera gave a positive result with end-titres ranging from 1 : 4 to 1 : 16, seven yielded end-titres below the specific threshold dilution (Table 5).

TABLE 5
End-titre ranges of sera from 25 patients with *T. b. gambiense* (Boma/Vindu-Zaire)

Sera		End-titres					
Drug-treated since	Number	Non-reactive	1 : 1	1 : 2	1 : 4	1 : 8	1 : 16
Untreated	4	1	0	1	0	2	0
4-9 months	21	0	1	6	7	5	2
Total	25	1	1	7	7	7	2

A third series of sera had been collected in Upper Volta from 23 patients, the majority of whom had been drug-treated. The 12 serum samples collected within 2 months after drug-treatment yielded end-titres ranging from 1 : 8 to 1 : 32. Eight out of 11 samples collected more than 2 months after chemotherapy showed titres of 1 : 4 or 1 : 8 whereas 3 reacted negatively (Table 6).

TABLE 6
End-titre ranges of sera from 23 patients with *T. b. gambiense* (Upper Volta)

Sera		End-titres				
Drug-treated since	Number	1 : 2	1 : 4	1 : 8	1 : 16	1 : 32
0 to 7 days	7	0	0	0	2	5
≤ 2 months	5	0	0	1	3	1
2 to 4 months	5	1	0	4	0	0
4 to 7 months	6	2	3	1	0	0
Total	23	3	3	6	5	6

Finally, sera of 31 untreated patients from Daloa and Bouaflé, Ivory Coast were tested. All serum samples yielded end-titres ranging from 1 : 4 to 1 : 32 (Table 7).

TABLE 7
End-titre ranges of sera from 31 patients with *T. b. gambiense* (Daloa/Bouaflé-Ivory Coast)

Sera		End-titres			
Origin	Number	1 : 4	1 : 8	1 : 16	1 : 32
Daloa	25	3	8	8	6
Bouaflé	6	0	1	4	1
Total	31	3	9	12	7

The overall results obtained with the 123 sera collected from trypanosomized patients before or within two months after drug-treatment, with the 247 sera from Africans and Caucasians free of trypanosomiasis and with the 54 sera from patients suffering from various parasitoses, are summarized in figure 2.

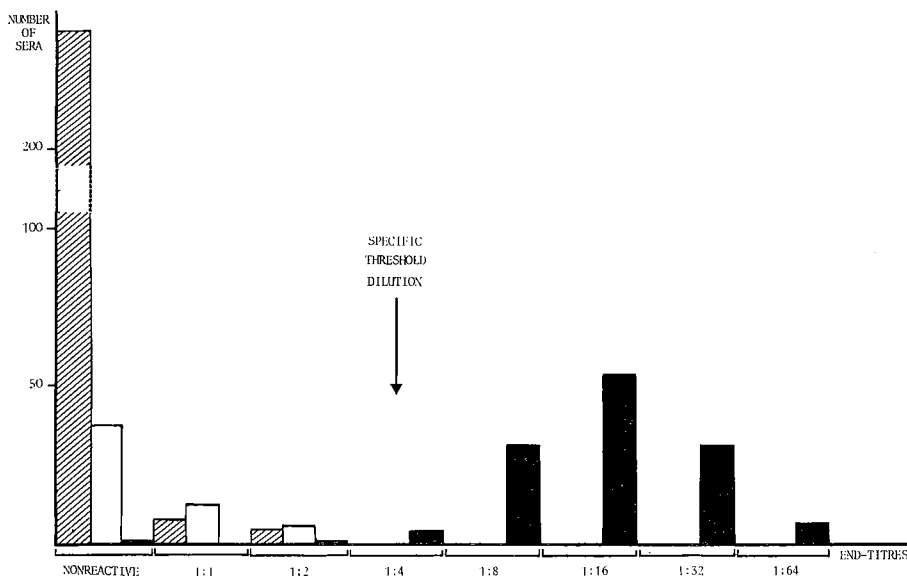


Figure 2

End-titre ranges of i) sera from 123 proven cases of *T.b.gambiense* collected before drug-treatment or within the next two months (■), ii) 247 sera of 56 Africans and 191 Caucasians free of trypanosomiasis (▨) and iii) sera from 54 patients with different parasitoses (□)

Discussion

Though it has been known for many years that relapsing infections with salivarian trypanosomes are characterized by the continual appearance of new variable antigen types (VATs) and by the production of corresponding host antibodies, the possibility of using variable antigens for serodiagnostic purposes has seldom been taken into consideration.

Recent research into antigenic variation and VAT repertoires revealed that some so-called predominant antigen types, appearing in an early stage of infection, are produced in common by taxonomically and geographically different isolates (stocks, demes) of the subgenus *Trypanozoon* (Gray and Luckins, 1976; Van Meirvenne *et al.*, 1977). This overlap of VAT repertoires suggested the potential interest of ubiquitous VATs for serodiagnostic tests.

Working with living trypanosomes of undefined antigen type, direct agglutination tests have occasionally been used for serodiagnosis of human and bovine trypanosomiasis (Soltys, 1957; Cunningham and Van Hoeve, 1964). Later on, Gray's experiments with clone populations have led to a better insight into the multiplicity of the VAT-specific agglutinating antibodies in sera of patients infected with *T. b. gambiense* (reviewed by Gray and Luckins, 1976).

The test described in the present paper is a rapid method for the detection of variant specific antibodies. Beside its diagnostic potential it also offers new possibilities for serodemological studies aiming at identification of VAT repertoires.

The C. A. T. T., carried out with VAT AnTat 8, proved sensitive for diagnosis of human *T. b. gambiense* trypanosomiasis in endemic areas of Zaïre, Upper Volta and Ivory Coast. The results suggest that antibody titres tend to decrease within a few months after drug-treatment. The specificity of the test needs further evaluation. Further comparative studies with a comprehensive set of VATs should reveal which ones are the most suitable for use in different foci of sleeping sickness.

By virtue of its simplicity and low cost price, the C. A. T. T. seems well suited for use in field conditions. Preliminary results suggest that the test could be performed on a drop of whole blood collected by finger-prick.

In its present form, the antigen retains full potency for more than 1 year when stored at -80°C , for at least 4 weeks at $+4^{\circ}\text{C}$ and for 24 hours at $+37^{\circ}\text{C}$. Longer storage at $+37^{\circ}\text{C}$ results in a progressive loss of reactivity. Attempts are now being made to increase the stability of the reagent by adding preservatives and/or by freeze-drying.

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Een kaart-agglutinatietest met gekleurde trypanosomen (C. A. T. T.) voor de serodiagnose van *T. b. gambiense* trypanosomiase.

Samenvatting — Er werd een kaart-agglutinatietest uitgewerkt voor de serodiagnose van menselijke *T. b. gambiense* trypanosomiase. Als antigeen werd een suspensie van gefixeerde en gekleurde trypanosomen van een welbepaald variabel antigeentype gebruikt. De waarde van de test werd nagegaan door titratie van sera afkomstig van 155 patiënten met *T. b. gambiense*, sera van 247 personen vrij van trypanosomiase en sera van 54 patiënten met diverse parasitosen.

Un test d'agglutination sur carte utilisant des trypanosomes colorés (C. A. T. T.) pour le sérodiagnostic de la trypanosomiase à *T. b. gambiense*.

Résumé — Un test d'agglutination sur carte pour le sérodiagnostic de la trypanosomiase humaine à *T. b. gambiense* a été développé. Une suspension de trypanosomes sanguicoles de type antigénique défini, fixés et colorés, a été employée comme antigène. Le test a été évalué en examinant des sérums provenant de 155 patients infectés par *T. b. gambiense*, de 247 individus non-trypanosés et de 54 patients avec différentes parasitoses.

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